Question ID: f5247e52

Question

$y = ax^2 - c$

In the equation above, a and c are positive constants. How many times does the graph of the equation above intersect the graph of the equation $y = a + c$ in the xy-plane?

Answer

A. Zero

B. One

C. Two

D. More than two

Question ID: 3b4b8831

Question

$$38x^2 = 38(9)$$

What is the negative solution to the given equation?

Question ID: 2d2ab76b

Question

$y = x^2 - 1$

$y = 3$

When the equations above are graphed in the xy-plane, what are the coordinates (x, y) of the points of intersection of the two graphs?

Answer

A. $(2,3)$

and $(-2,3)$

B. $(2,4)$

and $(-2,4)$

C. $(3,8)$

and $(-3,8)$

D. $(\sqrt{2},3)$

and $(-\sqrt{2},3)$

Question ID: 6acdcece

Question

$$b - 72 = \frac{x}{y}$$

The given equation relates the positive numbers b, x, and y. Which equation correctly expresses x in terms of b and y?

Answer

A. $x = \frac{b-72}{y}$

B. $x = by - 72$

C. $x = \frac{by-72}{y}$

D. $x = by - 72y$

Question ID: 09e5e4d3

Question

If $\frac{42}{x} = 7x$, what is the value of $7x^2$?

Answer

A. 6

B. 7

C. 42

D. 294

Question ID: 928498f3

Question

$6x^2+5x-7=0$

What are the solutions to the given equation?

Answer

A. $\frac{-5\pm\sqrt{25+168}}{12}$

B. $\frac{-6\pm\sqrt{25+168}}{12}$

C. $\frac{-5\pm\sqrt{36-168}}{12}$

D. $\frac{-6\pm\sqrt{36-168}}{12}$

Question ID: bef4b1c6

Question

$$\frac{55}{x+6} = x$$

What is the positive solution to the given equation?

Question ID: 876a731c

Question

$y = x^2$
$2y + 6 = 2(x + 3)$

If (x, y) is a solution of the system of equations above and x > 0, what is the value of xy ?

Answer

A. 1

B. 2

C. 3

D. 9

Question ID: 2f958af9

Question

$$v^2 = \frac{LT}{m}$$

The formula above expresses the square of the speed v of a wave moving along a string in terms of tension T, mass m, and length L of the string. What is T in terms of m, v, and L ?

Answer

A. $T = \frac{mv^2}{L}$

B. $T = \frac{m}{v^2 L}$

C. $T = \frac{mL}{v^2}$

D. $T = \frac{L}{mv^2}$

Question ID: 2683b5db

Question

$T = 0.01(P - 40{,}000)$

In a city, the property tax T, in dollars, is calculated using the formula above, where P is the value of the property, in dollars. Which of the following expresses the value of the property in terms of the property tax?

Answer

A. $P = 100T - 400$

B. $P = 100T + 400$

C. $P = 100T - 40{,}000$

D. $P = 100T + 40{,}000$

Question ID: 4b6f0a3f

Question

$$x^2 - 5x - 24 = 0$$

What is the sum of the solutions to the given equation?

Question ID: 87a3de81

Question

$x^2+x-12=0$

If a is a solution of the equation above and $a>0$, what is the value of a ?

Question ID: 0980fcdd

Question

$$x^2 = 6x + y$$
$$y = -6x + 36$$

A solution to the given system of equations is (x,y). Which of the following is a possible value of xy ?

Answer

A. 0

B. 6

C. 12

D. 36

Question ID: 364a2d25

Question

$$x + y = 17$$
$$xy = 72$$

If one solution to the system of equations above is (x, y), what is one possible value of x ?

Question ID: c77ef2fb

Question

Blood volume, V_B, in a human can be determined using the equation $V_B = \frac{V_P}{1-H}$, where V_P is the plasma volume and H is the hematocrit (the fraction of blood volume that is red blood cells). Which of the following correctly expresses the hematocrit in terms of the blood volume and the plasma volume?

Answer

A. $H = 1 - \frac{V_P}{V_B}$

B. $H = \frac{V_B}{V_P}$

C. $H = 1 + \frac{V_B}{V_P}$

D. $H = V_B - V_P$

Question ID: 895628b5

Question

$$y = (x - 2)(x + 4)$$

$$y = 6x - 12$$

Which ordered pair (x, y) is the solution to the given system of equations?

Answer

A. $(0, 2)$

B. $(-4, 2)$

C. $(2, 0)$

D. $(2, -4)$

Question ID: 29ed5d39

Question

$$p = 20 + \frac{16}{n}$$

The given equation relates the numbers p and n, where n is not equal to 0 and $p > 20$. Which equation correctly expresses n in terms of p?

Answer

A. $n = \frac{p-20}{16}$

B. $n = \frac{p}{16} + 20$

C. $n = \frac{p}{16} - 20$

D. $n = \frac{16}{p-20}$

Question ID: 630897df

Question

The speed of sound in dry air, v, can be modeled by the formula $v = 331.3 + 0.606T$, where T is the temperature in degrees Celsius and v is measured in meters per second. Which of the following correctly expresses T in terms of v ?

Answer

A. $T = \frac{v + 0.606}{331.3}$

B. $T = \frac{v - 0.606}{331.3}$

C. $T = \frac{v + 331.3}{0.606}$

D. $T = \frac{v - 331.3}{0.606}$

Question ID: a4f61d75

Question

$x^2 - ax + 12 = 0$

In the equation above, a is a constant and $a > 0$. If the equation has two integer solutions, what is a possible value of a ?

Question ID: 75a32330

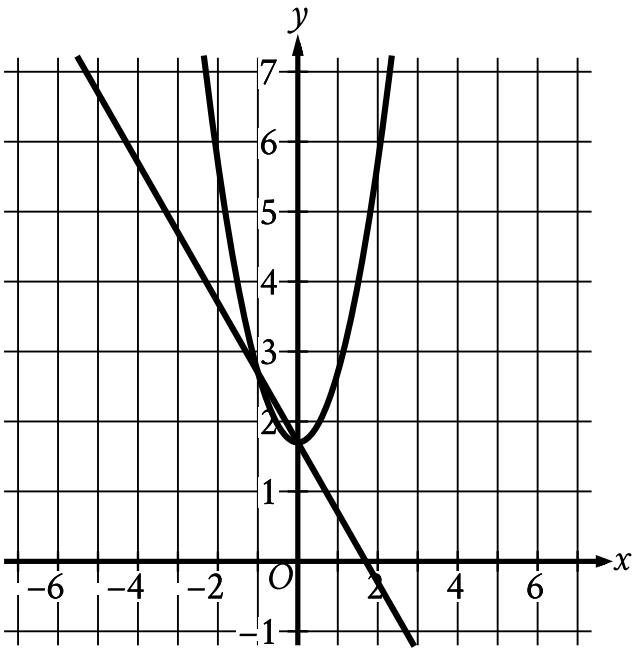
Question

$$y = x^2 + 1.7$$

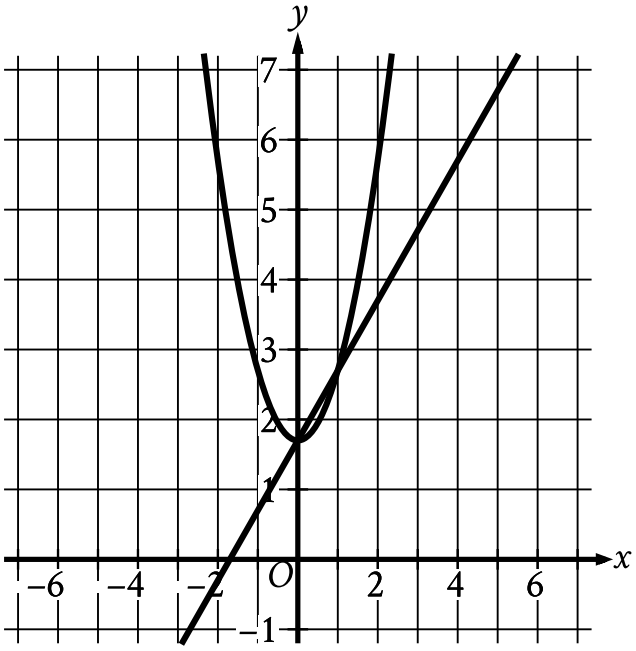
$$y = 1.7 - x$$

Which graph represents the given system of equations?

Answer

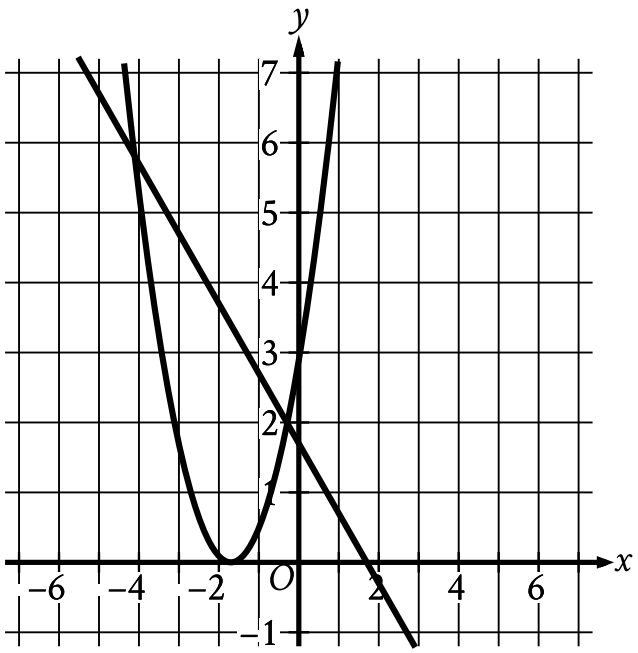


A.

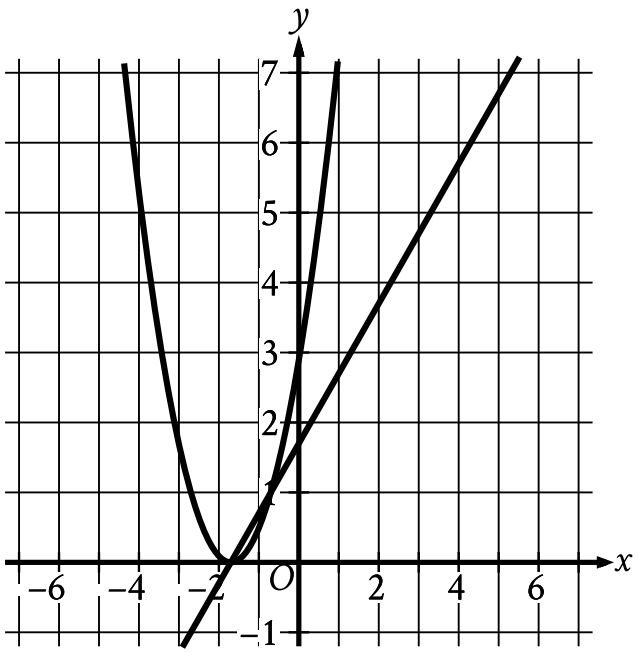


B.

C.



D.



Question ID: 802549ac

Question

$(x+2)(x+3)=(x-2)(x-3)+10$

Which of the following is a solution to the given equation?

Answer

A. 1

B. 0

C. -2

D. -5

Question ID: 11ccf3e1

Question

$$14j + 5k = m$$

The given equation relates the numbers j, k, and m. Which equation correctly expresses k in terms of j and m?

Answer

A. $k = \frac{m - 14j}{5}$

B. $k = \frac{1}{5}m - 14j$

C. $k = \frac{14j - m}{5}$

D. $k = 5m - 14j$

Question ID: 6e02cd78

Question

In the xy-plane, what is the y-coordinate of the point of intersection of the graphs of $y = (x-1)^2$ and $y = 2x-3$?

Question ID: 652054da

Question

An oceanographer uses the equation $s = \frac{3}{2}p$ to model the speed s, in knots, of an ocean wave, where p represents the period of the wave, in seconds. Which of the following represents the period of the wave in terms of the speed of the wave?

Answer

A. $p = \frac{2}{3}s$

B. $p = \frac{3}{2}s$

C. $p = \frac{2}{3} + s$

D. $p = \frac{3}{2} + s$

Question ID: fcdf87b7

Question

$y = x^2 - 4x + 4$

$y = 4 - x$

If the ordered pair (x, y) satisfies the system of equations above, what is one possible value of x ?

Question ID: b80d10d7

Question

$$\frac{2(x+1)}{x+5} = 1 - \frac{1}{x+5}$$

What is the solution to the equation above?

Answer

A. 0

B. 2

C. 3

D. 5

Question ID: 911383f2

Question

$(x-4)(x+2)(x-1)=0$

What is the product of the solutions to the given equation?

Answer

A. 8

B. 3

C. -3

D. -8

Question ID: 7f81d0c3

Question

$x^2 - x - 1 = 0$

What values satisfy the equation above?

Answer

A. $x = 1$ and $x = 2$

B. $x = -\frac{1}{2}$ and $x = \frac{3}{2}$

C. $x = \frac{1+\sqrt{5}}{2}$ and $x = \frac{1-\sqrt{5}}{2}$

D. $x = \frac{-1+\sqrt{5}}{2}$ and $x = \frac{-1-\sqrt{5}}{2}$

Question ID: d0a7871e

Question

$y = x + 1$

$y = x^2 + x$

If (x, y) is a solution to the system of equations above, which of the following could be the value of x ?

Answer

A. −1

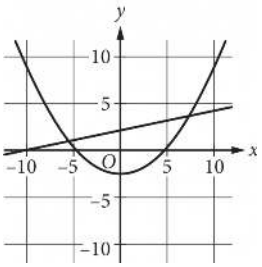
B. 0

C. 2

D. 3

Question ID: a5663025

Question



A system of equations consists of a quadratic equation and a linear equation. The equations in this system are graphed in the xy-plane above. How many solutions does this system have?

Answer

A. 0

B. 1

C. 2

D. 3